Premier University, Chittagong



Project Report

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1. Introduction

Heart disease is one of the leading causes of death worldwide and remains a major challenge for healthcare systems. Early detection and accurate prediction of heart disease risk are essential for timely medical intervention, reducing healthcare costs, and improving patient survival rates. As a result, developing reliable predictive systems for heart disease has become an important area of research in healthcare analytics. The main objective of this project is to develop a machine learning-based model that can classify whether a patient is at risk of heart disease using medical attributes such as age, blood pressure, cholesterol level, chest pain type, and other clinical features. This type of system can support doctors in making faster and more data-driven clinical decisions. In this study, several machine learning algorithms are applied and compared, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Neural Network models. The dataset is preprocessed using techniques such as handling missing values, feature encoding, and normalization to ensure better model performance. Unlike many previous studies that focus on a single model, this project performs a comparative analysis of multiple algorithms to identify the most effective approach for heart disease prediction. The experimental results show that Neural Networks and ensemble methods achieve higher performance than traditional models. In particular, the Neural Network model provides the best accuracy and AUC score, demonstrating strong predictive capability. Overall, the findings indicate that machine learning techniques can significantly improve early detection of heart disease and can be effectively used as a decision-support tool in clinical settings.

2. Problem Definition and Algorithm

2.1 Task Definition

This project addresses the problem of predicting heart disease risk levels using patient medical and lifestyle information. The task is formulated as a supervised multiclass classification problem**,** where a machine learning model is trained on historical patient data to predict the risk category for new, unseen cases.

Input (X):
The input consists of patient health records, including:

- **Numerical features:** age, resting blood pressure, serum cholesterol, maximum heart rate, ST depression (oldpeak), BMI, smoking status, and exercise level

- **Categorical features:** sex, chest pain type, resting electrocardiographic results, exercise-induced angina, and thalassemia condition

Output (Y):
The output is a categorical label representing heart disease risk level:

- Low Risk
- Medium Risk
- High Risk

Some models also generate probability scores for each class, which help in understanding prediction confidence.

Importance of the Problem:
Heart disease is a major cause of mortality worldwide, and early detection is critical for effective treatment. Predictive modeling helps identify high-risk patients in advance, supporting timely medical intervention. This problem is important in machine learning because it involves multiclass classification, mixed data types, and real-world noisy medical data, requiring both accuracy and reliability.

2.2 Algorithm Definition

To solve this problem, multiple machine learning algorithms are implemented and compared, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Neural Network (MLP).

Algorithm Workflow:

1. **Data Preprocessing**
    - Missing numerical values are replaced using median imputation
    - Missing categorical values are replaced using mode imputation
    - Numerical features are standardized using z-score normalization
    - Categorical features are converted using one-hot encoding
2. **Dataset Splitting**
    - The dataset is divided into 80% training and 20% testing data
3. **Model Training**
    - Each model is trained separately using the training dataset
4. **Prediction**
    - Models output class probabilities for each test sample

- Final prediction is selected using the highest probability (argmax)

5. **Evaluation**
   - Performance is measured using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC score

Pseudocode:

Input: Dataset D = (X, Y)
Output: Predicted heart disease risk class

1. Handle missing values:
- Replace numerical missing values with median
- Replace categorical missing values with mode

2. Preprocess features:
- Standardize numerical features
- One-hot encode categorical features

3. Split dataset:
- Train set (80%), Test set (20%)

4. Initialize models:
M = {Logistic Regression, Decision Tree, Random Forest,
SVM, KNN, Neural Network}

5. Train models:
For each model m in M:
m.fit(X_train, Y_train)

6. Predict:
For each model m in M:
P = m.predict_proba(X_test)
Y_pred = argmax(P)

7. Evaluate:
Compute accuracy, precision, recall, F1-score, ROC-AUC

8. Select best-performing model

Example Trace: Consider a patient with the following attributes

Age = 62, Sex = Female, Chest Pain = Typical Angina, Blood Pressure = 145, Cholesterol = 209, Max Heart Rate = 163, BMI = 20, Smoking = Yes, Exercise Level = Moderate

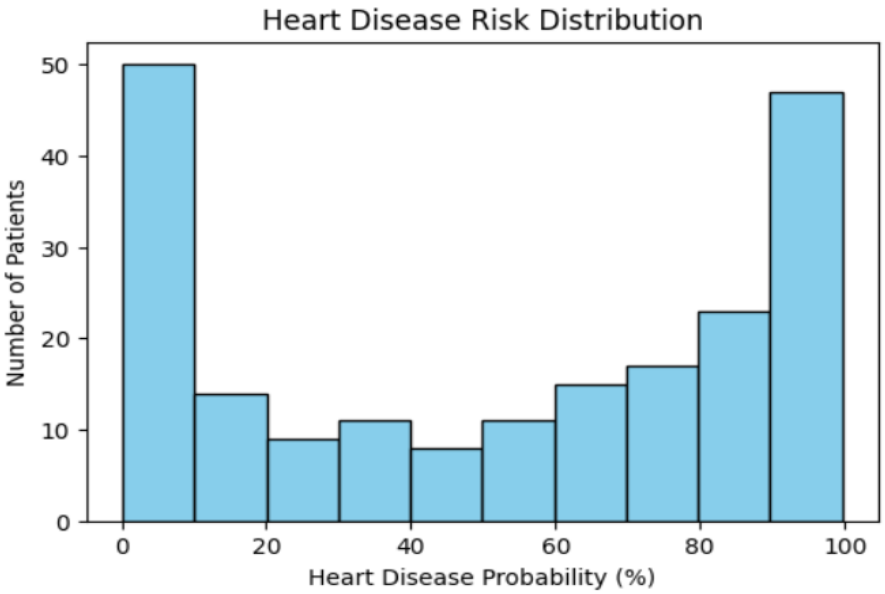
Step 1: Preprocessing
The raw input is converted into numerical format:

- Numerical values are standardized
- Categorical values are encoded

Step 2: Prediction
After feeding into trained models:

- Logistic Regression → Low: 0.04, Medium: 0.12, High: 0.84
- Random Forest → Low: 0.02, Medium: 0.08, High: 0.90

Step 3: Final Output
Both models predict: High Risk. This example shows how raw patient data is transformed into a structured prediction that can assist in early heart disease detection.



3. Related Work

Studies Based on the UCI Heart Disease Dataset

Many existing studies use the UCI Heart Disease dataset to develop machine learning models for heart disease classification. Common algorithms include Logistic Regression, Decision Trees, and Support Vector Machines (SVM). These works typically focus on building a single model and reporting its accuracy as the main performance measure. In contrast, our project evaluates multiple machine learning models within a unified experimental framework. We compare Logistic

Regression, Decision Tree, Random Forest, SVM, KNN, and Neural Networks under the same conditions to ensure a fair and consistent evaluation. This comparative approach provides a more complete understanding of model performance. It also helps identify the most effective algorithm for this dataset, where Neural Networks achieved the best results in terms of both accuracy and AUC.

Deep Learning-Based Heart Disease Prediction Models

Recent research has explored deep learning methods such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and hybrid architectures to improve prediction accuracy. These models are designed to capture complex relationships in medical data and often achieve strong performance. However, these approaches usually require large datasets and high computational resources. They are also less interpretable, which limits their practical use in healthcare settings. In this project, we use a simpler feed-forward Neural Network along with traditional machine learning models. This approach provides a balance between performance and interpretability while maintaining strong predictive accuracy.

Ensemble Learning Approaches (Random Forest and Boosting Methods)

Ensemble learning methods such as Random Forest, Gradient Boosting, and XGBoost are widely used in medical prediction tasks. These models are effective because they reduce overfitting and capture complex feature interactions. In our study, Random Forest is included as one of the baseline models and evaluated alongside other classifiers such as SVM, KNN, and Neural Networks. This allows direct comparison under identical conditions. The results show that although ensemble methods perform well, they are not always the best for this dataset. Neural Networks outperform Random Forest in both accuracy and AUC, providing clearer insight into the most suitable model.

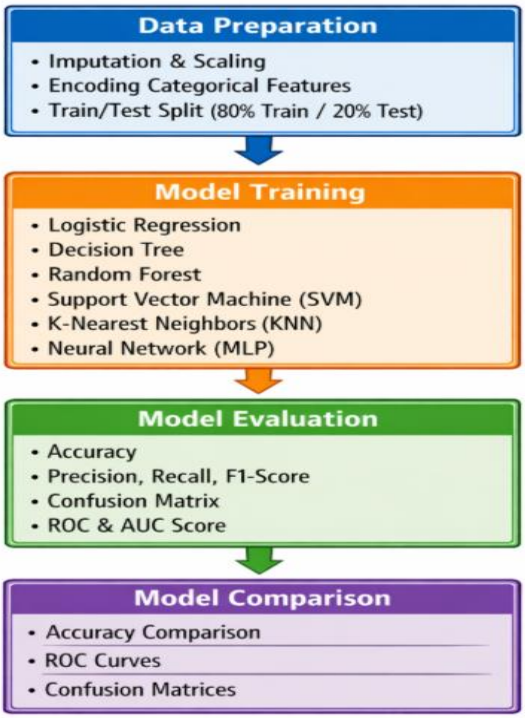
Support Vector Machine (SVM)-Based Studies

Support Vector Machines (SVM) with kernel functions have been widely used in heart disease prediction due to their ability to handle high-dimensional feature spaces and complex decision boundaries. In this project, SVM is evaluated alongside multiple models rather than being used as the sole classifier. This allows a fair comparison of its performance with other algorithms. The results indicate that while SVM performs strongly, Neural Networks still achieve better overall predictive performance. This highlights the importance of testing multiple models before selecting the best one for healthcare applications.

4. Experimental Evaluation

4.1 Methodology

The experimental evaluation is conducted to compare the performance of multiple machine learning models for heart disease risk prediction. The study follows a supervised learning framework using a labeled dataset, where each model is trained and tested under identical conditions.



Evaluation Criteria

Model performance is assessed using the following metrics:

- **Accuracy:** Measures the proportion of correctly classified instances.
- **Precision, Recall, and F1-score:** Evaluate class-wise prediction quality, particularly for imbalanced or overlapping classes.
- **Confusion Matrix:** Provides a detailed breakdown of correct and incorrect predictions for each risk category (Low, Medium, High).
- **ROC Curve and AUC Score:** Measure the ability of models to distinguish between classes across different decision thresholds.

Hypotheses

The experiments are designed to test the following assumptions:

1. Neural Networks achieve higher predictive performance than traditional machine learning models.
2. Logistic Regression and SVM serve as strong baseline models for structured medical data.
3. The Medium Risk class is more difficult to classify due to overlapping feature distributions with Low and High Risk categories.

Variables

- **Independent Variables:** Patient attributes such as age, cholesterol level, blood pressure, ECG results, chest pain type, smoking status, and exercise level.
- **Dependent Variable:** Heart disease risk level (Low, Medium, High).

Dataset and Experimental Setup

The dataset consists of structured patient medical records and is split into:

- **Training set:** 80%
- **Testing set:** 20%

This split ensures that models are trained on sufficient data while being evaluated on unseen samples, allowing fair comparison and generalization assessment.

Performance Data Collection

For each model, the following outputs are recorded:

- Overall accuracy
- Confusion matrix
- Classification report (precision, recall, F1-score)
- ROC curves and AUC scores

Model Comparison

The following models are evaluated under the same experimental conditions:

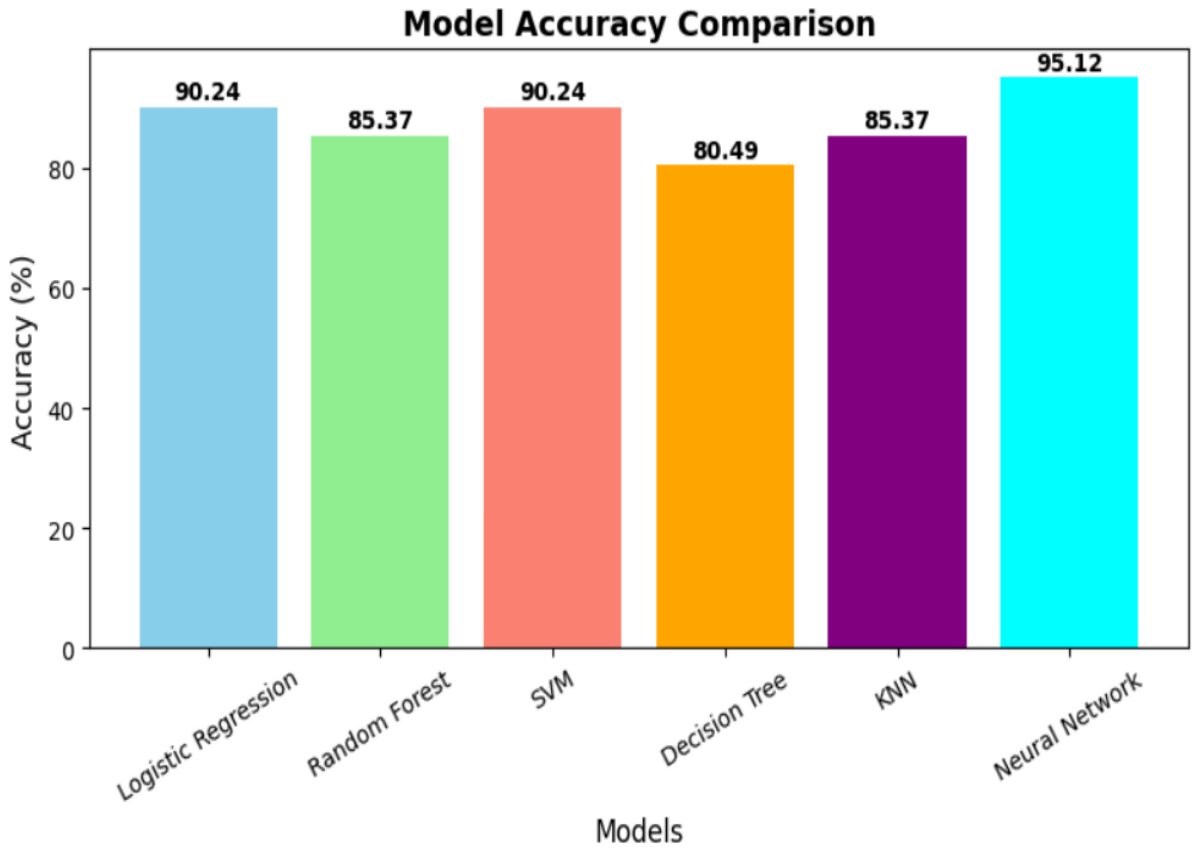
- Logistic Regression
- Decision Tree
- Random Forest

- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Neural Network (MLP)

Results are presented using:

- Accuracy comparison bar charts
- ROC curves
- Confusion matrix heatmaps

4.2 Results



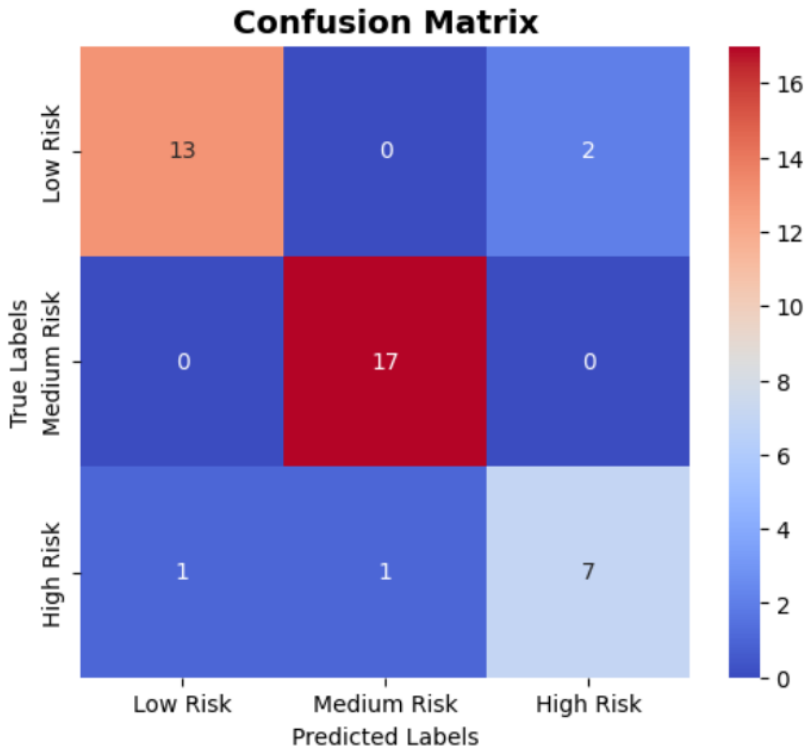
The experimental results reveal clear performance differences among the evaluated models.

Model Accuracy Comparison

- Neural Network: **95.12%**

- Logistic Regression: **90.24%**
- SVM: **90.24%**
- Random Forest: **85.37%**
- KNN: **85.37%**
- Decision Tree: **80.49%**

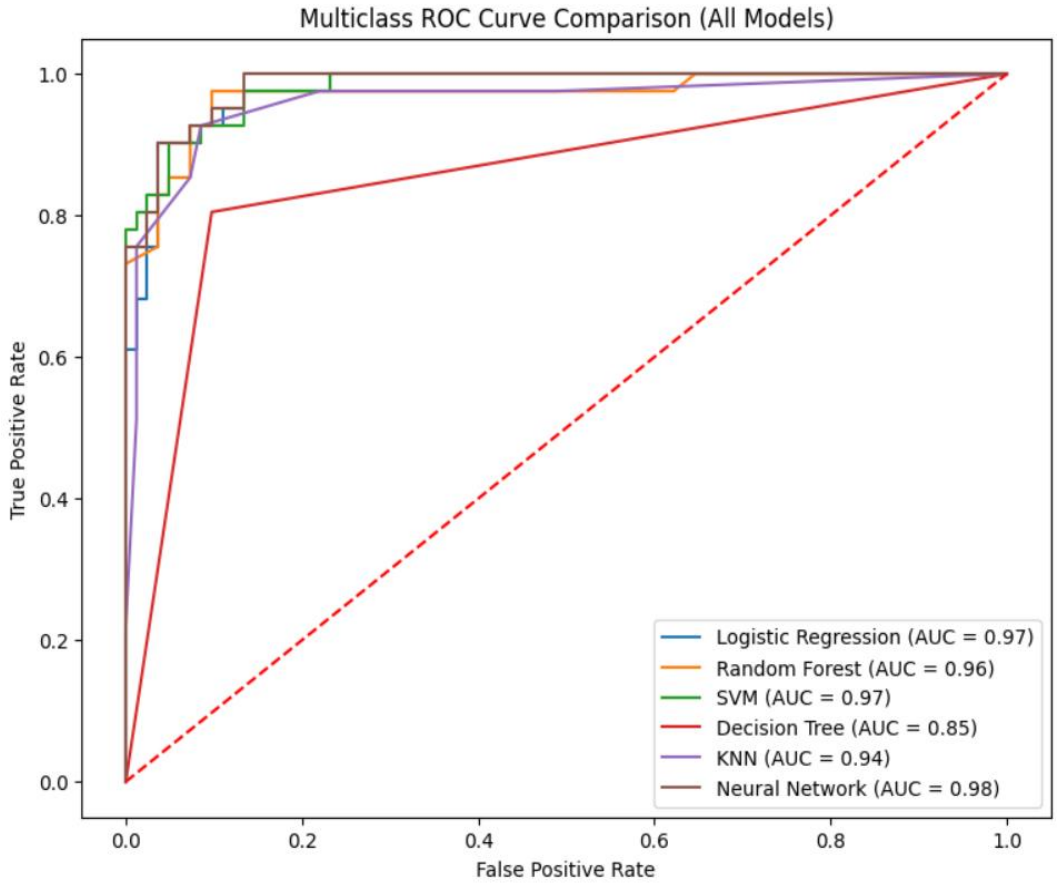
The Neural Network achieves the highest accuracy, demonstrating its ability to capture complex nonlinear relationships in the dataset.



Confusion Matrix Analysis

[[13 0 2] → High Risk
[0 17 0] → Low Risk
[1 1 7]] → Medium Risk

- **Low Risk:** Classified with high accuracy and strong recall.
- **High Risk:** Well identified with minimal misclassification.
- **Medium Risk:** More difficult to classify due to overlapping features with other classes.



ROC-AUC Performance

- Neural Network: **0.98**
- Logistic Regression: **0.97**
- SVM: **0.97**
- Random Forest: **0.96**
- KNN: **0.94**
- Decision Tree: **0.85**

The ROC results confirm that Neural Networks provide the best overall class separation ability.

Graphical Analysis

The following visualizations are used to support result interpretation:

- Bar chart comparing model accuracies
- ROC curves for all models
- Confusion matrix heatmaps
- Distribution plots of predicted risk categories

4.3 Discussion

The results support the initial hypotheses and provide meaningful insights into model behavior.

Key Findings

- The Neural Network outperforms all other models, confirming its strength in modeling nonlinear medical relationships.
- Logistic Regression and SVM perform competitively, indicating that the dataset contains partially linearly separable patterns.
- Decision Tree shows the weakest performance due to overfitting and limited generalization ability.

Strengths of Models

- **Neural Network:** Highest accuracy and strong generalization capability.
- **Logistic Regression:** Highly interpretable and stable baseline.
- **SVM:** Strong decision boundary modeling capability.
- **Random Forest:** Robust to noise and feature interactions.

Limitations

- Medium Risk classification remains challenging due to feature overlap with other classes.
- Simpler models struggle to capture complex nonlinear relationships in the data.

Interpretation of Results

The superior performance of Neural Networks is due to their ability to learn complex nonlinear interactions among medical features such as cholesterol levels, ECG patterns, and exercise response. In contrast, Logistic Regression and SVM perform well because parts of the dataset exhibit near-linear separability.

5. Future Work

Although the proposed system achieves strong performance in heart disease risk prediction, several limitations remain. Addressing these shortcomings can further improve model accuracy, robustness, and real-world applicability.

Shortcoming 1: Limited Dataset Size

- **Issue:**
  The current dataset contains a relatively small number of samples, which limits the model's ability to generalize to unseen real-world cases. It also increases the risk of overfitting, especially for complex models like Neural Networks.
- **Proposed Enhancement:**
  Future work should use larger and more diverse datasets collected from multiple hospitals or combine publicly available datasets such as different UCI Heart Disease sources. This would improve model generalization and reduce variance.

Shortcoming 2: Difficulty in Predicting Medium Risk Cases

- **Issue:**
  The model performs less effectively in predicting Medium Risk patients due to overlapping feature patterns with Low and High Risk classes. This makes class boundaries less clear.
- **Proposed Enhancement:**
  Techniques such as SMOTE (Synthetic Minority Oversampling Technique)**,** class-weighted learning, or ensemble-based hybrid models can be applied to improve class balance and decision boundary separation.

Shortcoming 3: Limited Model Interpretability

- **Issue:**
  Although Neural Networks provide high accuracy, they function as black-box models, making it difficult to explain predictions in clinical environments.
- **Proposed Enhancement:**
  Explainable AI methods such as SHAP and LIME can be used to interpret model predictions and identify the most influential medical features. This would improve trust and adoption in healthcare applications.

Shortcoming 4: Lack of Temporal Data

- **Issue:**
  The dataset represents only a single snapshot of patient health, whereas heart disease risk can change over time.
- **Proposed Enhancement:**
  Future systems should incorporate time-based data such as longitudinal medical records or wearable sensor data. Models like RNNs or LSTMs can be used to capture temporal patterns and disease progression.

6. Conclusion

This project explored the use of machine learning techniques for predicting heart disease risk levels based on patient medical and lifestyle data. A complete workflow was developed, including data preprocessing, model training, evaluation, and comparison of multiple classification algorithms. The results show that machine learning models perform well in predicting heart disease risk. Among all the tested models, the Neural Network (Multilayer Perceptron) achieved the highest performance with 95.12% accuracy and an AUC score of 0.98. It consistently outperformed Logistic Regression and Support Vector Machine, which both achieved around 90.24% accuracy. Random Forest and K-Nearest Neighbors showed moderate performance at approximately 85% accuracy, while the Decision Tree model performed the lowest at 80.49% accuracy due to overfitting and weaker generalization. The study contributes by building a complete machine learning pipeline for heart disease prediction, comparing six different classification models under the same conditions, identifying Medium Risk cases as the most difficult class due to overlapping feature patterns, and highlighting the trade-off between accuracy and interpretability, where Neural Networks performed best while Logistic Regression provided better explainability. Overall, this project demonstrates that machine learning can effectively support early detection of heart disease. While advanced models such as Neural Networks provide the highest accuracy, simpler models like Logistic Regression and Decision Trees remain important for interpretable and practical clinical decision-making.

Bibliography

- Breiman, L. (2001). *Random forests*. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324 Introduces the Random Forest algorithm widely used for classification and prediction tasks.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297. https://doi.org/10.1007/BF00994018 Presents Support Vector Machines (SVM), a key method for classification in machine learning.
- Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. Nature, 323(6088), 533–536. https://doi.org/10.1038/323533a0 Introduces the backpropagation algorithm used in training neural networks.
- Dua, D., & Graff, C. (2019). UCI Machine Learning Repository: Heart Disease Dataset. University of California, Irvine. https://archive.ics.uci.edu/ml/index.php
  A standard dataset widely used for heart disease prediction research.
- World Health Organization. (2024). Cardiovascular diseases (CVDs) fact sheet. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)
  Provides global information on cardiovascular disease and its impact.